

AI-Powered Placement Assist & Student Readiness Platform

Overview

A full-stack AI-powered placement assistance platform that helps students prepare for placements, interviews, communication rounds, and technical assessments using multi-modal AI modules including NLP, Voice Processing, Computer Vision, OCR, and Video Analytics.

1■■■ TEXT MODULE — Smart Placement Task Intelligence

- Classifies Gmail and WhatsApp placement messages automatically.
- Extracts deadlines, interview schedules, coding rounds, and assessments.
- Creates intelligent to-do lists based on urgency and dates.
- AI revision assistant recommends important topics before tests/interviews.
- Prioritizes DSA, aptitude, and interview preparation tasks.

Technologies: spaCy, Transformers, Sentence Transformers, FastAPI

2■■■ VOICE MODULE — AI Communication & Interview Coach

- Detects filler words like 'uh', 'ah', 'umm'.
- Analyzes speaking speed and confidence.
- Provides live feedback such as 'slow down' or 'speak clearly'.
- Detects nervousness, hesitation, and panic during speech.
- Converts lectures and YouTube videos into summarized notes.

Technologies: Whisper, librosa, Speech Recognition APIs

3■■■ IMAGE MODULE — Document Verification & Profile Intelligence

- OCR-based extraction from marksheets and certificates.
- Detects fake certificates using logo/watermark matching.
- Analyzes LinkedIn/profile photos for professionalism.
- Converts handwritten notes into searchable digital text.
- Automatically stores extracted student data into database.

Technologies: Tesseract OCR, OpenCV, EasyOCR

4■■■ VIDEO MODULE — HR Interview & Body Language Analyzer

- Tracks eye contact during mock interviews.
- Analyzes posture, gestures, and confidence.

- Detects nervous body movement and fidgeting.
- Evaluates facial expressions and communication behavior.
- Generates interview readiness and confidence scores.

Technologies: MediaPipe, OpenCV, Pose Detection Models

■ ■ Suggested Tech Stack

Component	Technology
Frontend	React / Streamlit
Backend	FastAPI / Flask
Database	MongoDB / Firebase
AI Models	Transformers, Whisper, OpenCV
Deployment	Docker / Render / AWS

Project Value & Impact

This project combines NLP, Speech AI, OCR, Computer Vision, and Video Analytics into one integrated placement assistance platform. It helps students improve communication, placement preparation, interview confidence, and task management while also reducing manual work for placement cells. The project demonstrates strong real-world relevance, scalability, and industry-level AI integration.